

VIOLA Exposure Time Calculator (VIOLA ETC)

User Manual

App User Edition

VIOLA ETC Team

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<https://viola-etc.streamlit.app/>

Scope of this document. This manual explains how to use the VIOLA ETC web application to estimate signal-to-noise ratio (SNR) performance and inspect related diagnostic outputs. It is intended for users who run the ETC through the hosted Streamlit app. Mathematical formulation, software architecture, and supporting data files are described in the Technical Documentation.

For questions, feedback, or bug reports, contact sirinrat@narit.or.th.

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1 Overview

1.1 What VIOLA ETC is

The VIOLA Exposure Time Calculator (VIOLA ETC) is a Python-based modeling tool that predicts the signal-to-noise ratio (SNR) achievable for the VIPA-based Instrument for an Oxygen-Loaded Atmosphere (VIOLA) spectrograph under user-defined observing conditions.

VIOLA is a high-throughput, ultra-high-resolution near-infrared spectrograph ($R \approx 300,000$), with the prototype covering the J (1.1–1.3 μm) and H (1.5–1.7 μm) bands, designed to support the detection of molecular signatures in exoplanet atmospheres. The ETC also supports calculations for K band (1.9–2.5 μm), while K-band coverage on the VIOLA instrument itself is currently under consideration.

The ETC computes an end-to-end photon budget from source to detector, incorporating telescope and instrument throughput, telluric atmospheric transmission, and key noise contributions including sky emission, detector noise, and thermal background. An optional module estimates the detectability of planetary molecular features using a precomputed five-class template grid and the High-Resolution Cross-Correlation Spectroscopy (HRCCS) matched-filter framework, under simplified assumptions.

The ETC is available as a Streamlit-based web application providing an interactive interface for exploring instrument performance, observation planning, and science case exploration.

1.2 Typical use cases

- Estimate SNR versus wavelength for a target with a given magnitude and observing setup.
- Compare observing strategies (e.g., exposure time, number of frames, nod subtraction settings, telescope aperture).
- Identify the dominant noise regime (stellar photon noise vs. background vs. detector terms).
- Assess the detectability of planetary molecular features for a chosen planet class and line contrast using the first-order molecular detectability module.

1.3 What the ETC does not claim

The ETC outputs are intended for **forecasting and trade studies**. They are not a substitute for full end-to-end simulations, injection–recovery tests, or a detection claim for exoplanet atmospheres. The molecular detectability module is a first-order estimator that does not model line blending, explicit telluric overlap, stellar activity, planet orbital motion, or atmospheric dynamics. Any molecule-related outputs should be interpreted as approximate feasibility estimates under the stated assumptions.

2 Access and Requirements

2.1 How to access

The VIOLA ETC is available as a hosted web application:

<https://viola-etc.streamlit.app/>

Source code and technical documentation are available at:

<https://github.com/sirinrats/viola-etc>

2.2 User requirements

No special software is required beyond a standard desktop or laptop web browser. The app is not yet validated on mobile phones or tablets. If you encounter display or functionality issues, please report them.

2.3 Availability notes

The app is hosted on Streamlit Cloud. If the page does not load or returns an error, refresh the page and re-run the calculation. Performance may depend on server load.

3 Quick Start: 60-second workflow

1. Open the app: <https://viola-etc.streamlit.app/>
2. Set **Target** inputs: observing band, magnitude system, stellar magnitude, effective temperature, and radial velocity (applied to both the stellar spectrum and the planetary molecular templates).
3. Optionally expand **Target (advanced)** to change the SED model or enable the molecular detectability module.
4. Set **Observing conditions**: telescope aperture, exposure time, number of frames, target altitude, PWV, seeing, and nod subtraction.
5. Click **Run ETC**.
6. Inspect the outputs across the five tabs:
 - **Results**: median SNR and SNR versus wavelength, plus a magnitude-sweep sensitivity curve.
 - **Sky Model**: atmospheric transmission and sky background components.
 - **Signal & Noise**: breakdown of the signal and noise budget per resolution element.
 - **Molecules**: first-order molecular detectability estimate for the selected planet class.
 - **Debug**: input snapshot for reproducibility and issue reporting.

4 User Interface Guide

The interface is organized into three panels. The **Target** panel defines the astrophysical source; an expandable **Target (advanced)** sub-panel provides additional spectral energy distribution (SED) and molecular detection controls. The **Observing conditions** panel sets the telescope and integration parameters. The **Outputs** panel displays the results across five tabs.

4.1 Target panel

Observing band

Select the spectral band for the SNR calculation. The band determines the wavelength range used throughout the ETC and selects the corresponding sky-model region:

- **J band**: 1.10–1.30 μm
- **H band**: 1.50–1.70 μm
- **K band**: 1.90–2.50 μm

Magnitude system and stellar magnitude

Select the photometric system (**Vega** or **AB**) and enter the apparent stellar magnitude in the selected band. The ETC scales the source model to match this magnitude at the band centre.

Stellar effective temperature

Enter the stellar effective temperature T_{eff} in Kelvin. This controls the spectral shape of the SED model used to distribute flux across the band. The overall flux level is anchored to the input magnitude regardless of T_{eff} .

Stellar radial velocity

Enter the stellar radial velocity v_r in km s^{-1} (range: $\pm 2000 \text{ km s}^{-1}$). A positive value denotes recession (redshift). The velocity shifts the stellar spectrum (and the planet atmosphere template) relative to the telluric rest frame, which is relevant for molecular cross-correlation analyses. For a blackbody SED, the shift affects only the continuum shape via rest-frame wavelength evaluation, no spectral lines are displaced. For PHOENIX-NewEra, it displaces the stellar absorption features relative to telluric lines.

4.2 Target (advanced) sub-panel

Expand the **Target (advanced)** section to access additional settings.

Stellar SED model

Choose the SED model:

- **Blackbody:** a Planck function scaled to the input magnitude. Allowed T_{eff} range: 2000–40,000 K.
- **PHOENIX-NewEra:** synthetic stellar atmosphere spectra from Hauschildt et al. (2025) at fixed $\log g = 4.5$ and solar abundances. Grid coverage: $T_{\text{eff}} = 2300\text{--}12,000 \text{ K}$. If the requested T_{eff} falls outside this range, the ETC automatically falls back to a blackbody.

Molecular detection

Enable the **Molecular detection** checkbox to activate the first-order template-based detectability module. When enabled, additional controls appear:

Planet class. Select the planet atmosphere class. This determines which precomputed spectral template set is loaded. Five classes are available, with the available molecules listed in Table 1.

Table 1: Planet classes and available molecular templates.

Planet class	Available molecules
Temperate terrestrial ($\sim 250 \text{ K}$, rocky)	CH_4 , CO , CO_2 , H_2O , O_2
Temperate sub-Neptune ($\sim 270 \text{ K}$, H_2 -dominated)	CH_4 , CO_2 , H_2O , NH_3
Warm sub-Neptune ($\sim 600 \text{ K}$)	CH_4 , CO , CO_2 , H_2O , NH_3
Hot Jupiter ($\sim 1500 \text{ K}$)	CH_4 , CO , CO_2 , H_2O
Ultra-hot Jupiter ($\sim 2500 \text{ K}$)	CO , CO_2 , H_2O , OH

Templates for temperate planet classes are computed with the NASA Planetary Spectrum Generator (PSG). Templates for hot planet classes use petitRADTRANS. All templates are precomputed at fixed atmospheric assumptions and should be interpreted as approximate, feasibility-level inputs.

Planet-to-star line contrast. Enter the assumed fractional line depth of the planet signal relative to the stellar continuum (C_{line} ; dimensionless). This parameter scales the planetary signal in the detectability estimate and reflects the strength of absorption or emission features in the planet atmosphere.

Detection thresholds.

- **Minimum SNR per resolution element** — lines at wavelengths with stellar SNR below this value are excluded (they fall in deep telluric gaps or suppressed continuum regions).
- **Minimum number of lines** — if fewer lines pass the selection thresholds, the detection metrics are reported as N/A.
- **Minimum line depth (normalized)** — lines whose template prominence (as a fraction of the in-band dynamic range) falls below this value are excluded.
- **Detrending efficiency (0–1)** — fraction of the planet signal amplitude that survives telluric removal, stellar-line subtraction, and other pipeline detrending steps. Users may adjust this parameter to explore optimistic or conservative signal recovery scenarios appropriate to their observing strategy and data reduction pipeline.
- **Required detection significance (σ)** — the target detection significance used to estimate the number of observing nights needed.

4.3 Observing conditions panel

Telescope aperture

Select the telescope diameter: **2 m** (Wendelstein Observatory Fraunhofer Telescope, LMU Munich, Germany) or **8 m** (one unit telescope of the ESO Very Large Telescope (VLT), Paranal, Chile). The aperture sets the collecting area and scales both the target signal and sky background.

Exposure time per frame and number of frames

Set the exposure time per frame t_{exp} (seconds) and the total number of frames N_{exp} . The total on-source integration time is:

$$t_{\text{total}} = N_{\text{exp}} \times t_{\text{exp}}$$

In the molecular detectability module, the full N_{exp} -frame observation is treated as one allocated observing unit (displayed as one transit in the current app version).

Target altitude and precipitable water vapor (PWV)

These two inputs select the sky model used by the ETC. The ETC draws from a precomputed grid of 45 ESO SkyCalc FITS files covering:

- **Target altitude:** 30°, 45°, 60°, 75°, 90°
- **PWV:** 0.5, 1.0, 1.5, 2.5, 3.5, 5.0, 7.5, 10.0, 20.0 mm

For any user-specified values, the nearest grid point is selected automatically. Lower altitude increases airmass and reduces atmospheric transmission. Higher PWV increases water-vapour absorption, particularly in the H and K bands.

Seeing full width at half maximum (FWHM)

The seeing FWHM (arcsec) determines the fraction of stellar light that enters the spectrograph through the slit or fiber entrance via an error-function throughput model. The seeing significantly larger than the entrance width value will reduce the detected signal.

A–B nod subtraction

When enabled, the ETC assumes an A–B nod-on-slit observing mode. The telescope alternates between two nod positions along the slit. Pairs of frames are differenced during data reduction to suppress sky and thermal background. Because both nod positions contribute independent background shot noise, the background Poisson variance is doubled (pair factor $f_{\text{pair}} = 2$), which lowers the predicted SNR compared to the no-nod case. In practice, nod subtraction is used to control background residuals and instrumental systematics, so this SNR penalty should be weighed against its practical benefits.

5 Understanding Outputs

After clicking **Run ETC**, results are displayed across five tabs in the output panel.

5.1 Results tab

The primary performance outputs:

- **Median SNR per resolution element** — a single summary number for the current configuration.
- **SNR vs. wavelength** — the per-resolution-element SNR across the selected band. The array can be downloaded as a CSV file.
- **Median SNR vs. magnitude** — a sensitivity curve showing how the median SNR scales with target magnitude under fixed observing assumptions. The sweep range and step size can be customized above the Run ETC button.

5.2 Sky Model tab

Visualizes the sky-model inputs across the selected band:

- Atmospheric transmission (total, and decomposed into molecular absorption, ozone, Rayleigh scattering, and Mie scattering components)
- Sky emission components from the SkyCalc model (zodiacal light, OH airglow, molecular thermal emission, scattered moonlight, scattered starlight, and airglow continuum)

The displayed model corresponds to the nearest grid point to the user-specified altitude and PWV. It is intended for feasibility comparisons rather than site-specific predictions.

5.3 Signal & Noise tab

Shows the signal and main noise contributions across the band:

- Target signal per resolution element
- Target signal per pixel
- Sky background per resolution element
- Instrument thermal background per resolution element
- Dark current per resolution element

This tab is useful for identifying the dominant noise term at each wavelength (photon-noise limited, background limited, or read-noise limited).

5.4 Molecules tab

Available when the **Molecular detection** option is enabled. For each molecule in the selected planet class, the tab displays:

- The molecular template spectrum and the predicted SNR spectrum overlaid on the same axes, with selected lines highlighted
- The number of usable lines passing both the prominence and SNR thresholds
- The estimated detection significance per observing night:

$$\text{SNR}_{\text{night}} = p_{\text{detrend}} \cdot C_{\text{line}} \left[\sum_{j \in \mathcal{L}} \text{SNR}_{\text{res},j}^2 \right]^{1/2}$$

- The estimated number of observing nights required to reach the user-specified detection significance:

$$N_{\text{nights}} = \left(\frac{\sigma_{\text{det}}}{\text{SNR}_{\text{night}}} \right)^2$$

If fewer lines pass the thresholds than the minimum number of lines, detection metrics are reported as N/A. These outputs are first-order estimates and should not be interpreted as detection claims.

5.5 Debug tab

Provides a snapshot of the run configuration for reproducibility and issue reporting:

- All input parameter values and their data types
- The last-run input set used to generate the current results
- Result metadata for diagnosing unexpected behavior

When reporting a problem, include a screenshot of any error message and the relevant content from this tab.

6 Version History

Version	Key changes
v0.3	First Streamlit web application.
v0.4	Added PHOENIX-NewEra SED model. SkyCalc grid with altitude/PWV selection. Configurable telescope aperture (2 m / 8 m).
v0.5	Added stellar radial velocity input. Upgraded molecular detection to a five-class precomputed planet template grid (PSG + petitRADTRANS). Extended SkyCalc grid PWV range to 20 mm. Updated detection formula to use per-line SNR quadrature sum. Bug fixes: read-noise variance, N_{exp} double-counting, detrending factor placement.

7 Troubleshooting and FAQ

The app page does not load.

Refresh the browser. If the problem persists contact the ETC team.

The SNR output is very low or zero.

Check that the target magnitude is within a reasonable range for the chosen telescope and exposure setup. Very faint targets or very short exposures can produce near-zero SNR. Inspect the Signal & Noise tab to identify which noise term dominates.

Molecular detection shows N/A for all molecules.

Fewer lines passed the selection thresholds than the minimum required. Try reducing the minimum number of lines, the minimum SNR threshold, or the minimum line depth. You can also try a different planet class.

PHOENIX-NewEra falls back to Blackbody.

The requested T_{eff} is outside the PHOENIX-NewEra grid range (2300–12,000 K). Either adjust T_{eff} or use the Blackbody model directly.

I want to report a bug or unexpected result.

Include a screenshot of any error message and the full contents of the Debug tab in your report. Contact: sirinrat@narit.or.th.